Tool Analysis Document

GitTruck@Duck: Interactive Time Range Selection in Hierarchy-Oriented Polymetric Visualization of Git Repository Evolution

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Installation and Usage

GitTruck@Duck is exceptionally easy to install and use, following a "private-by-design" philosophy that requires no complex cloud setup or database configuration. To begin, the user simply navigates to a Git repository or a folder containing several repositories within their terminal. The installation is performed globally by running the command `npm install -g git-truck@latest`, which installs or updates the tool to its most recent version. Once initiated, the application is launched by executing the command `git truck`, which automatically triggers the interface to open in the system's default web browser. Because the tool runs entirely locally, it analyzes the `.git` directory of whichever repository it is currently in and serves the visualization via a local web server. This local-first approach ensures that the tool is highly portable, running seamlessly in any repository you need on your machine without the risk of uploading sensitive source code to external servers.

Benefits and Findings

The tool offers significant benefits for both individual developers and project managers through its unique visual approach to repository evolution. The standout "Duck" feature provides an interactive time range selector that allows users to isolate specific periods of development, which is invaluable for pinpointing exactly when a codebase underwent rapid expansion or when specific technical debt was introduced. By mapping metrics such as commit frequency to file size and top contributors to color, the tool provides an immediate visual heatmap for polymetric hotspot detection, allowing users to find high-churn files that may require refactoring. Furthermore, the visualization highlights knowledge distribution and potential knowledge silos. By shifting the time slider, one can observe if a critical module has transitioned from being team-maintained to being managed by a single individual, identifying potential bus-factor risks. Unlike flat list views, GitTruck preserves the folder hierarchy using treemaps or bubble charts to provide structural insight, allowing users to understand how different modules grow in relation to the project's overall architecture.

Screen Captures

Paper: *GitTruck@Duck: Interactive Time Range Selection in Hierarchy-Oriented Polymetric Visualization of Git Repository Evolution*
Authors: Adrian Hoff, Thomas Hoffmann Kilbak, Leonel Merino, Mircea Lungu

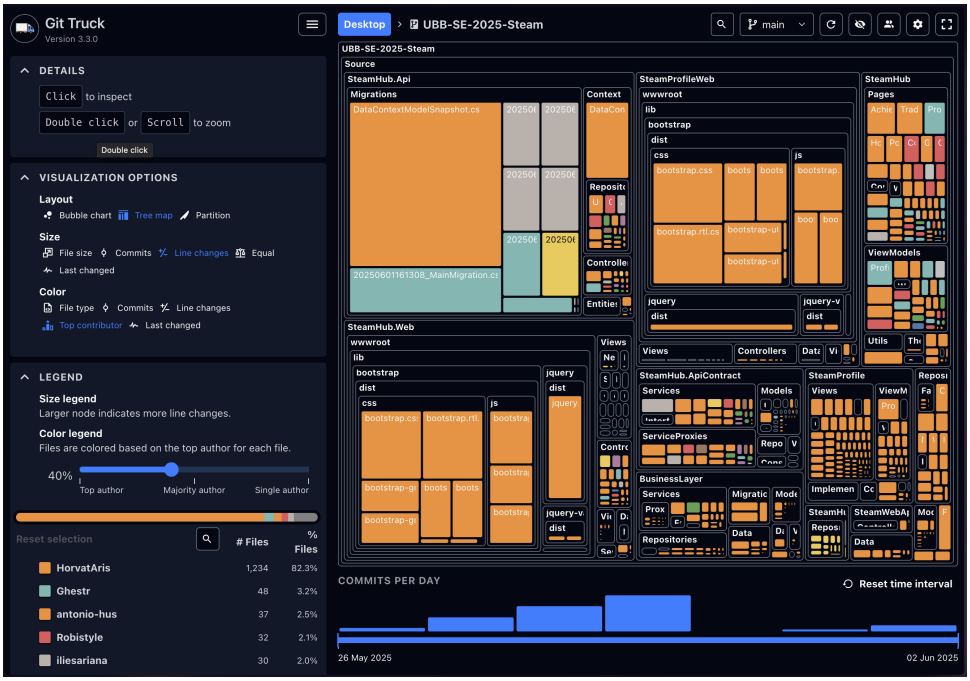

Figure 1: Main Dashboard showing the Treemap layout of a repository with files colored by Top Contributor

Figure 2: The Time Range Selector at the bottom of the UI being adjusted to a specific historical window

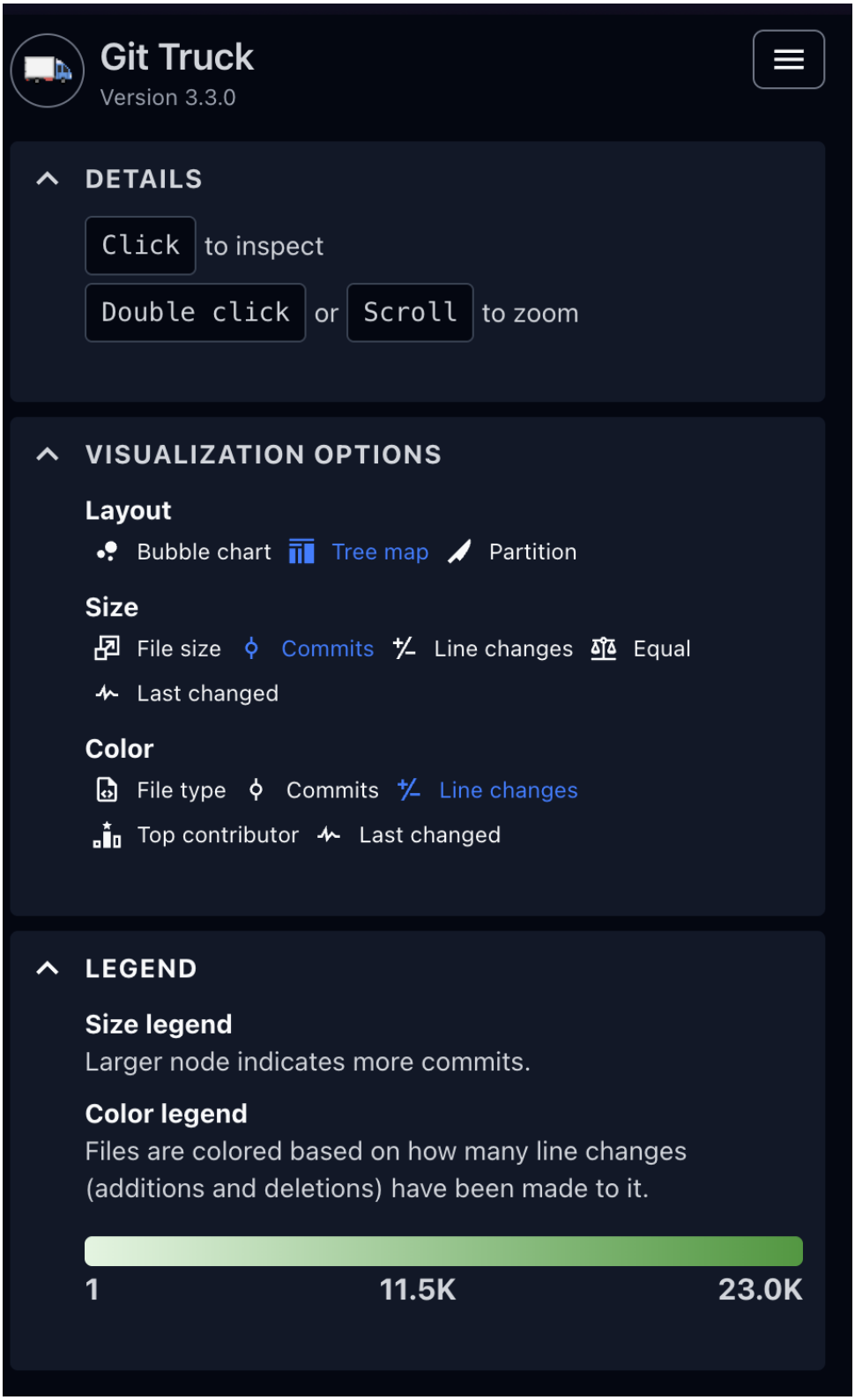

Figure 3: The Configuration Side Panel showing the selection of metrics for Size and Color encoding

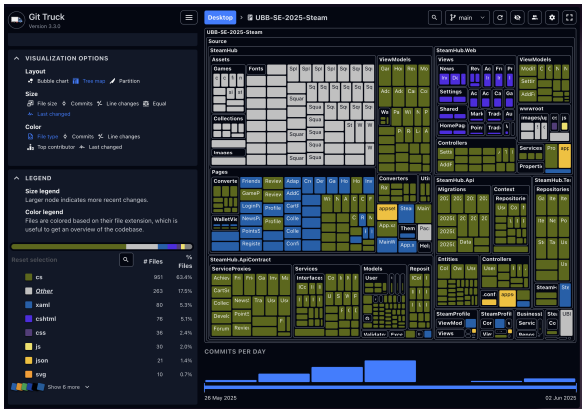

Figure 4: Initial repository state showing high activity and file distribution in the early selected time range.

Figure 5: Updated view after adjusting the time range, highlighting shifts in commit density and file modifications.